

PAPER_1396 — UQFF Resolution of the St. Petersburg Paradox (CLOSED — Dispatch Whitepaper)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Foundational / Decision Theory (CLOSED) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "st_petersburg"})` **Calculator surface:** `calculate_paradox({"paradox": "st_petersburg"})` **Closure helper:** `_l96_uqff_axiom_st_petersburg_closure()`

The Paradox

Coin-toss game: pay $\$2^n$ where n is the first heads. Expected payout = $\sum (1/2)^n \times 2^n = \infty$. Yet no rational player would bid more than $\sim \$20$. Classical resolutions invoke log-utility or risk aversion — all parametric.

UQFF Closed Identity

$D_{crit!} = 26! = 4.03 \times 10^{26}$ finite expectation bound

$D_{crit!} = 26! \approx 4 \times 10^{26}$ is the structural finite bound on infinite geometric sums in the UQFF vacuum, the same truncation that bounds eternal inflation (PAPER_597 / `eternal_inflation_closure`). The expected payout truncates at finite n_{max} where the SCm vacuum cannot sustain larger occupation modes.

Physical Interpretation

The paradox dissolves because the infinite geometric sum is bounded by $D_{crit!}$ — a structural integer-primitive truncation. The ‘rational bid’ emerges from the D_{crit} -bounded expectation, not from utility curves.

Live Calculator Output

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "st_petersburg"})["value"]
```

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_st_petersburg_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` $\rightarrow$ `_l96_uqff_axiom_st_petersburg_closure`
- Dispatch keys: `st_petersburg`, `st_petersburg_paradox`
- Related foundational papers: `PAPER_646` (`F_U_Bi_i` / `F_U=1` ledger), `PAPER_597` (negative-time dual existence), `PAPER_1183` (paradox consolidation), `PAPER_1156` (cosmology suite), `PAPER_1376` (Banach-Tarski parallel structural closure).

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